PH45130

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Quantum Gravitation

Lecture 5

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Massive Dispersion

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$E^2 = p^2 c_m^2 + m_g^2 c_m^4$$

Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$v_{\mathrm{g}}/c_m \approx 1 - m_{\mathrm{g}}^2 c_m^4/(2E^2)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$m_g \lesssim 10^{-32} eV/c^2$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 5; M32-r5 methods; One-sided Limits Standard OSL-8